Sri Lanka Institute of Information Technology

IT2130 – Operating Systems and System Administration
Year 2, Semester 2- 2026

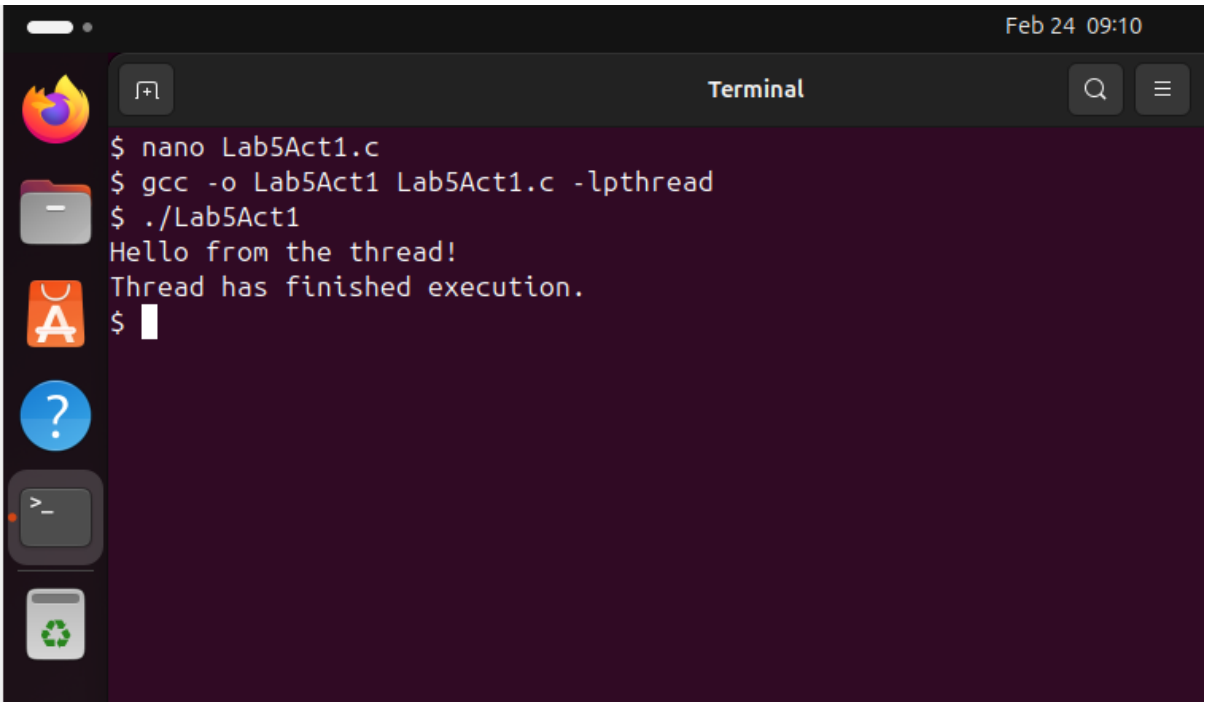
Practical Answer Submission

Practical Sheet No: 05

Student ID	IT24102555
Student Name	Weerathunga B.A
Campus/ Center name	SLIIT-Malabe
Specialization	Data Science
Batch No	Y2.S2-WD-DS-01.02

Activity 1

a)



b) printf ()

pthread_create ()

pthread_join ()

pthread_exit ()

perror ()

main ()

c)The program creates a new thread using pthread_create ()

The thread runs the function print_message

Thant function prints “hello from the thread!”

pthread_join () makes the main thread wait until the created thread finishes.

After that, main prints “Thread has finished execution”

Activity 2

Feb 24 09:46

Terminal

GNU nano 7.2 Lab5Act2.c

```c
#include <stdio.h>
#include <pthread.h>

void*print_thread(void*arg){
   int thread_num =*(int*)arg;
   printf("Thread %d is running \n",thread_num);
    return NULL;
}

int main(){
   pthread_t thread[3];
   int thread_nums[3];

  for(int i=0;i<3;i++){
     thread_nums[i]=i+1;
     pthread_create(&thread[i],NULL,print_thread,&thread_nums[i]);
}
  for(int i=0;i<3;i++){
  pthread_join(thread[i],NULL);
}
   return 0;
}
```

Feb 24 09:45

Terminal

```
$ nano Lab5Act2.c
$ gcc -o Lab5Act2 Lab5Act2.c -lpthread
$ ./Lab5Act2
Thread 1 is running
Thread 2 is running
Thread 3 is running
$
```

Activity 3

Feb 24 09:56

Terminal

GNU nano 7.2 Lab5Act3.c

```c
#include <stdio.h>
#include <pthread.h>

typedef struct{
    int a;
    int b;
} Numbers;

void*calculate(void*arg){
   Numbers*nums =(Numbers*)arg;
   int sum = nums->a +nums->b;
   printf("Sum = %d\n",sum);
   return NULL;
}
int main(){
    pthread_t t;
    Numbers n;
    n.a =10;
    n.b =20;
pthread_create(&t,NULL,calculate,&n);
pthread_join(t,NULL);
 return 0;
}
```

Feb 24 09:55

Terminal

```
$ nano Lab5Act3.c
$ gcc -o Lab5Act3 Lab5Act3.c -lpthread
$ ./Lab5Act3
Sum = 30
$
```

a) The typedef struct is used to:
   - Create a structure named Numbers
   - Store two integer values (a and b)
   - Pass multiple values to thread using a single argument

b) Pthread_create () requires the thread function to accept a void*argument
   It allows passing any type of data to the thread.
   Inside the thread, it must be typecast to the correct type.

C) arg is a void * pointer

It is typecast to Numbers *

Now nums can access a and b using:

nums - >a

nums ->b

<u>Activity 4</u>

Activity 04

```c
#include <stdio.h>
#include <pthread.h>

typedef struct {
    int a;
    int b;
} Numbers;

void* calculate_sum(void* arg) {
    Numbers* nums = (Numbers*)arg;
    int sum = nums->a + nums->b;
    printf("Sum = %d\n", sum);
    return NULL;
}

void* calculate_product(void* arg) {
    Numbers* nums = (Numbers*)arg;
    int product = nums->a * nums->b;
    printf("Product = %d\n", product);
    return NULL;
}

void* array_sum(void* arg) {
    int* arr = (int*)arg;
    int total = 0;
    for(int i = 0; i < 4; i++) {
        total += arr[i];
    }
```

```
  GNU nano 7.2                                    Activity4.c
    }
    printf("Array Total Sum = %d\n", total);
    return NULL;
}

int main() {
    pthread_t t1, t2, t3;
    Numbers n;
    int arr[4];

    printf("Enter two numbers: ");
    scanf("%d %d", &n.a, &n.b);

    printf("Enter 4 numbers for the array: ");
    for(int i = 0; i < 4; i++) {
        scanf("%d", &arr[i]);
    }

    pthread_create(&t1, NULL, calculate_sum, &n);
    pthread_create(&t2, NULL, calculate_product, &n);
    pthread_create(&t3, NULL, array_sum, arr);

    pthread_join(t1, NULL);
    pthread_join(t2, NULL);
    pthread_join(t3, NULL);

    return 0;
}
```

```
$ nano Activity4.c
$ gcc -oLab5act4 Activity4.c
$ ./Lab5act4
Enter two numbers: 5 4
Enter 4 numbers for the array: 1 2 3 4
Sum = 9
Product = 20
Array Total Sum = 10
$
```